

Finite-size–corrected synchronization in a city-scale bus network: crossover, not demonstrated criticality

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Bus bunching is often interpreted as phase synchronization whose onset occurs when passenger demand exceeds a critical value. We test this on 13.85 million GPS fixes and 6.79 million boarding records from 20 routes in Niterói, Brazil. The raw first Kuramoto–Daido order parameter is not comparable across observations: under independent random phases its baseline scales approximately as $0.886/\sqrt{N}$, while the median observed fleet contains only six buses. We validate same- N Monte Carlo nulls against Kluyver’s exact planar-random-walk distribution and compare five normalizations, two phase definitions, and four null models. Corrected order is positively associated with realized boardings per active bus, but only one of three predeclared criticality signatures passes; a free-exponent scaling collapse is optimizer-boundary and start-grid sensitive and is therefore not identified. The appropriate nearest-neighbour Newell–Potts model predicts instability for every positive load rather than a positive threshold, although its measured load parameter does not predict which hourly cells amplify. Spatial position provides the dominant empirical reinterpretation: 51.37% of active bus-hours lie in the outer 10% of a route, terminal order is nearly one, and interior corrected order is -0.189 . The whole-route load coefficient is 0.0376 , whereas the interior coefficient is 0.0011 ($p = 0.933$); direct control for terminal fraction attenuates the whole-route coefficient by 44%. Shared-corridor coherence also falls toward zero after controlling shared-segment speed. Thus an apparently Kuramoto-like field signal is not evidence of a city-scale critical transition: it is substantially shaped by finite-fleet bias, boundary occupancy, inherited operational irregularity, and common road-state shocks.

1. INTRODUCTION

A delayed bus encounters passengers accumulated over a longer headway, dwells longer, and loses more time; the following bus encounters fewer passengers and catches it. This positive feedback, formalized by Newell and Potts, makes bus bunching

a natural candidate for the language of synchronization. In a Kuramoto interpretation, buses are oscillators, position along a route is phase, and increasing passenger load strengthens an interaction until collective phase locking appears. A threshold-like curve of an order parameter against demand can therefore look like a phase transition.

That inference requires more than a rising curve. A field observable must have the same baseline when the number of observed oscillators changes; its phase coordinate must advance uniformly under uncoupled motion; its boundaries must not create order by construction; and the proposed transition must show physical signatures beyond a fitted breakpoint. Real bus systems violate all four conditions: fleets are small and vary by hour, speed is non-uniform, routes are open and terminate, and interactions are local and directed rather than mean-field.

We use the NetMob 2026 Niterói data to separate an apparent Kuramoto signal from what the data identify physically. Our contribution is threefold. First, we put finite- N correction on an exact random-walk foundation and test it across estimators and nulls. Second, we distinguish a descriptive load crossover from criticality using predeclared physical gates and a falsifiable Newell–Potts comparator. Third, we make route position a first-class variable, revealing that whole-route order combines terminal queueing with in-service spacing. The resulting conclusion is not that buses never interact, but that raw coherence alone cannot identify collective critical dynamics.

2. DATA AND IDENTIFICATION

The common observation window is 11–31 March 2026. It contains 13,850,467 telemetry fixes from 527 hardware IDs and 6,790,681 ticket records from 556 operator vehicle numbers, together with GTFS feeds, route geometry, stops, and hourly weather. The two vehicle identifier sets have zero overlap and no released crosswalk. Boardings therefore cannot be assigned to individual GPS vehicles. We join the feeds only by normalized route and local hour, after explicit aliases such as 62B→62.

The analysis unit is line–direction–date–hour. Because tickets do not identify direction, both directions receive the same route-hour boarding numerator but retain direction-specific active-bus denominators,

$$\lambda_{\ell dt} = \frac{B_{\ell t}}{N_{\ell dt}}. \quad (1)$$

We call λ realized boardings per active bus. It is not exogenous demand, passenger load on a particular vehicle, or a causal

Table 1. Raw-to-analysis ledger. Exclusion flags can overlap and must not be summed.

Quantity	Recomputed value
GPS fixes	13,850,467
Ticket rows	6,790,681
Selected routes / directions	20 / 40
Layover flags	3,920
Sparse-midpoint flags	11,326
High-residual flags	948
Active bus-hours	69,466
Hourly cells	11,641
Analysis-eligible cells	10,135

treatment: supply N responds to operating conditions and to expected demand.

3. GEOMETRY, PHASE, AND CELLS

A. Continuous route geometry

For each of 20 high-ridership routes, GPS points are projected onto a validated route polyline. For point $\mathbf{p}$ and segment $\mathbf{a}$ to $\mathbf{b}$, with $\mathbf{v} = \mathbf{b} - \mathbf{a}$,

$$t^* = \text{clip}_{[0,1]} \frac{(\mathbf{p} - \mathbf{a})^\top \mathbf{v}}{\mathbf{v}^\top \mathbf{v}}, \quad \hat{\mathbf{p}} = \mathbf{a} + t^* \mathbf{v}. \quad (2)$$

The closest segment gives both the residual distance and cumulative arc position s .

MultiLineString members are globally ordered and oriented by dynamic programming, minimizing first the maximum connector and then the total connector distance. A static route is rejected if the best remaining connector exceeds 250 m. This validation is consequential for route 49.2: its continuous length is 14.54 km and its largest member connector is 7.84 m; concatenating stored members would instead create a 3.62 km synthetic link and a 21.28 km route while leaving perpendicular GPS residuals deceptively small.

B. Bus-hour representation and exclusions

Each line-direction-vehicle-hour is represented by the eligible GPS fix closest to the clock-hour midpoint (HH:30), not by an average over the trajectory. A bus-hour is excluded if its nearest fix is more than 20 minutes from the midpoint or its median matching residual exceeds 150 m. A terminal layover is flagged when observation duration is at least 10 minutes, at least 75% of samples lie in the outer 5% of the route, and arc span is no more than 150 m. Table 1 records the recomputed counts.

C. Two phase coordinates

The conventional arc phase is

$$\phi_{\text{arc}} = 2\pi s / L. \quad (3)$$

It assumes constant free-running speed, which buses do not have. As a control we construct a travel-time phase. Vehicle-day traces are separated into trips at terminal resets; crossing times at 5%, 10%, 20%, ..., 90%, and 95% of route length yield median

inter-section travel times from at least 20 runs. Their cumulative profile $T(s)$ gives

$$\phi_{\text{time}} = 2\pi T(s) / T_{\text{cycle}}. \quad (4)$$

Profiles exist for all 40 line-directions and cover all active bus-hours. This coordinate controls unequal typical speed; it does not force measured order to decrease, because it stretches slow segments and compresses fast ones.

For a cell with N active buses, the Kuramoto–Daido harmonics are

$$Z_m = \frac{1}{N} \sum_{j=1}^N e^{im\phi_j}, \quad r_m = |Z_m|, \quad m = 1, 2, 3. \quad (5)$$

The complete rebuild reproduces all 11,641 cell keys and fleet sizes. Corrected estimators are undefined for $N = 1$, where $r_1 = 1$ deterministically.

4. FINITE-SIZE MEASUREMENT THEORY

Under independent uniform phases, Nr_1 is the endpoint length of an N -step planar random walk. Kluyver's exact CDF is

$$P(r_1 \leq x) = Nx \int_0^\infty J_1(Nxt) J_0(t)^N dt, \quad (6)$$

with $E[r_1^2] = 1/N$ exactly and $E[r_1] = 2/\pi$ at $N = 2$. The familiar $E[r_1] \simeq \sqrt{\pi}/(2\sqrt{N}) = 0.8862/\sqrt{N}$ is only the Rayleigh large- N approximation. Here the median cell has $N = 6$ and the lower quartile $N = 4$.

We use 5,000 Monte Carlo draws for the Phase-2 base table and 20,000 for the full estimator/null sweep and zone analyses. The implementation is tested against Kluyver's CDF and exact moments. The main reported excess is

$$r_{\text{exc}} = \frac{r_1 - \mu_N}{1 - \mu_N}, \quad (7)$$

where μ_N is the same- N null mean. For robustness we also compute

$$z = \frac{r_1 - \mu_N}{\sigma_N}, \quad \text{PPC} = \frac{Nr_1^2 - 1}{N - 1}, \quad Z_R = Nr_1^2, \quad (8)$$

and a probit-transformed probability-integral score. PPC equals the mean pairwise cosine over distinct bus pairs and is unbiased for the squared population resultant under i.i.d. sampling.

Four nulls preserve progressively more structure: independent uniform phase; speed-weighted arc occupancy; empirical occupancy resampling; and line-direction-conditioned sampling without replacement that breaks simultaneity. Speed and uniform are identical under time phase, leaving seven executable phase-null specifications and 35 estimator-specification contrasts.

5. MODEL DISCRIMINATION

The mean-field Kuramoto model assumes constant natural phase velocity, attractive all-to-all coupling, a closed ring, a fixed frequency distribution, and a large ensemble. A bus route has local directed interaction, terminal resets, changing membership, and median $N = 6$. Computing a Kuramoto order parameter therefore does not establish Kuramoto dynamics.

For the nearest-neighbour Newell–Potts mechanism, if dwell increment is $kh_{n,j}$, headway perturbations evolve as

$$h_{n,j+1} = (1 + k)h_{n,j} - kh_{n-1,j}. \quad (9)$$

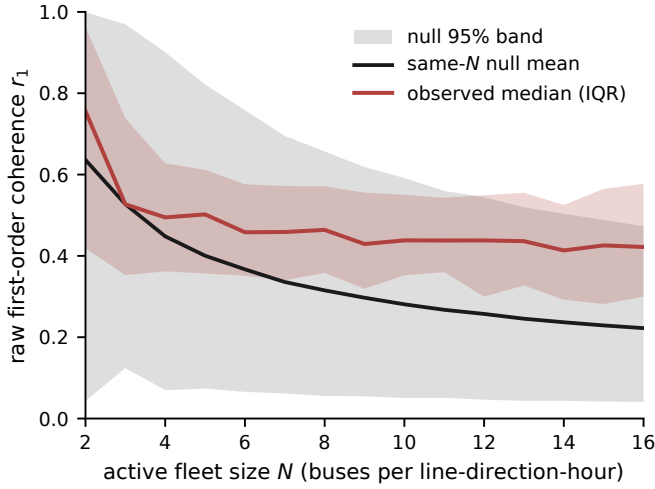


Fig. 1. Raw order has a moving finite-size baseline. Observed medians should be compared with the same- N null, not across fleet sizes directly. The grey region is the null 95% interval; the red region is the observed interquartile range.

Substituting $h_{n,j} = A^j e^{i\theta n}$ yields

$$A(\theta) = (1 + k) - k e^{-i\theta}, \quad (10)$$

$$|A|^2 = (1 + k)^2 - 2k(1 + k) \cos \theta + k^2. \quad (11)$$

Thus $|A(0)| = 1$, while $|A(\theta)| > 1$ for every $k > 0$ and nonzero θ . The idealized model has no positive load threshold: instability accelerates continuously with k . This is a theoretical comparator, not proof that observed cells obey the model.

6. RESULTS

A. Raw and corrected weekend comparisons

In raw arc phase, weekend minus weekday r_1 is $+0.045$ (95% CI $[0.036, 0.055]$), while mean fleet size falls from 7.99 to 4.47. Matching exactly within N reduces the raw contrast to $+0.014$. Under the arc-uniform correction the marginal contrast reverses, $\Delta r_{\text{exc}} = -0.068$ ($[-0.087, -0.049]$), whereas the exact- N -matched contrast is $+0.012$ ($[-0.009, +0.031]$).

Across all seven specifications, 30 of 35 marginal corrected contrasts are negative and all 35 matched point estimates are positive; 24 matched confidence intervals exclude zero. This is not a contradiction. The marginal estimand includes the weekend supply pathway through N ; the matched estimand asks for a day-type association at the same active fleet size. Neither is a causal weekend treatment effect.

B. Load response: crossover without demonstrated criticality

A random-intercept model with line-direction effects, standardized N , time-of-day harmonics, weekend, rain, and a load-rain interaction gives a standardized load coefficient of 0.04256 ($p < 10^{-15}$). This is an association with realized load. On the same fixed-effect frame, a piecewise curve has AIC 10308.21, compared with 10334.61 for linear and 10396.36 for sigmoid. Its descriptive scale is $\lambda_c = 135.74$ boardings per active bus.

We predeclare three signatures of criticality: an interior susceptibility peak, bimodality or coexistence near the candidate scale, and stability of the apparent scale across N . The first two fail and only N -stability passes ($|\text{corr}(\lambda_{\text{peak}}, N)| = 0.282$).

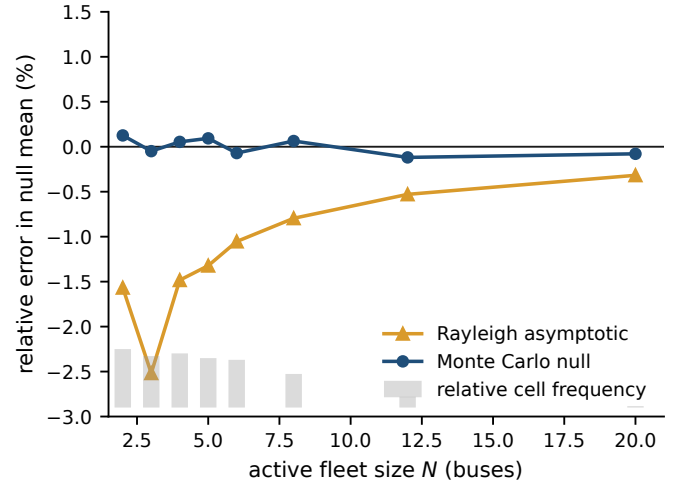


Fig. 2. The Rayleigh approximation is biased where most cells lie, whereas the simulated null agrees with the exact Kluyver mean within Monte Carlo error. Bars show the relative frequency of the displayed N values, not an additional response variable.

Table 2. Weekend-weekday contrasts, arc phase and uniform null.

Estimator	Marginal	Matched on N
Raw r_1	$+0.045$	$+0.014$
r_{exc}	-0.068	$+0.012$
z	-0.458	$+0.133$
PPC	-0.043	$+0.003$
Z_R	-0.580	$+0.187$
probit PIT	-0.399	$+0.112$

The verdict is one of three signatures, consistent with a smooth crossover and no demonstrated criticality.

A free-exponent collapse $r = N^{-b} F((\lambda - \lambda_c) N^a)$ is non-diagnostic. In the new raw rebuild, the whole-route narrow-start solution improves the objective by 3.43 (permutation $p = 0.167$), while a broad start gives 10.01 at a different exponent-boundary solution. The interior fit is likewise start-sensitive and boundary-limited. We therefore set `collapse_identified=false` and interpret neither exponents nor permutation values as transition evidence.

The corrected Daido harmonics are nearly flat: arc-phase means are 0.157, 0.159, and 0.149 for $m = 1, 2, 3$; time-phase means are 0.271, 0.293, and 0.275. A broad single von Mises cluster tuned at $m = 1$ decays toward zero at higher harmonics. Flat harmonics therefore describe compact clumps, not the emergence of one smooth mean-field phase density.

C. Route position is a first-class variable

Even after layover exclusion, 27.19% of active bus-hours lie within 2% of a route end and an additional 24.18% lie between 2% and 10%; only 48.63% are in the 10–90% interior. Median matching residuals are similar in the three zones (4.24, 3.87, and 4.14 m), so the boundary mass is not a projection artifact.

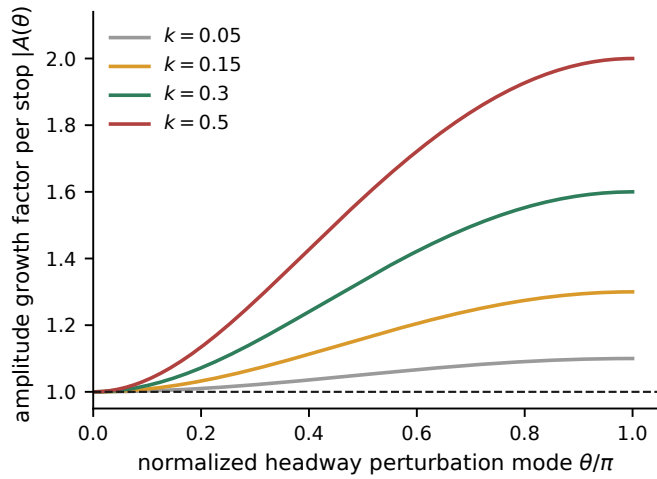


Fig. 3. Newell–Potts growth factors. The uniform-shift mode $\theta = 0$ is neutral; every nonzero mode grows for every positive k . Hence the idealized bus-following theory predicts no positive load threshold.

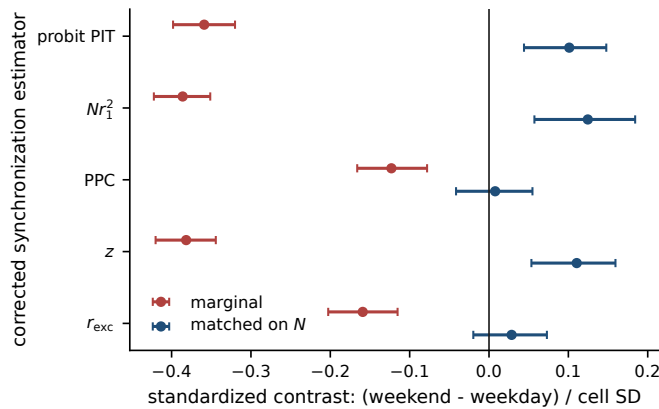


Fig. 4. Marginal and exact- N -matched weekend contrasts, scaled by each estimator's empirical standard deviation. All estimators change direction with the estimand. The comparison is about conditioning on fleet size, not choosing the estimator that gives a preferred sign.

Order parameters are recomputed inside each zone, with a new within-zone N . Terminal cells have mean $r_{\text{exc}} = 0.9974$, near-terminal cells 0.9186, and interior cells -0.1887 . Negative interior excess means more even spacing than the specified same- N uniform null; it is consistent with a splay-like state but does not by itself prove successful dispatch control.

The standardized whole-route load coefficient in a line-direction fixed-effect model is 0.03758 (95% CI [0.01097, 0.06420], $p = 0.00565$). No zone has a detectable within-zone load coefficient: the interior estimate is 0.00111 ([$-0.02460, 0.02683$], $p = 0.9325$). Directly adding terminal fraction to the whole-route model reduces the load coefficient to 0.02106, a 44% attenuation; terminal fraction itself has coefficient 1.2386 ($p \approx 8.2 \times 10^{-182}$). Thus the whole-route association is substantially compositional, not proven to be purely compositional.

Trimming 2% from both ends changes arc-phase mean r_{exc} from 0.155 to -0.120 ; at 5% it is -0.223 . More severe trimming eventually removes genuine route and reduces median N to three, so 2–10% is the informative range. Each trim is a differ-

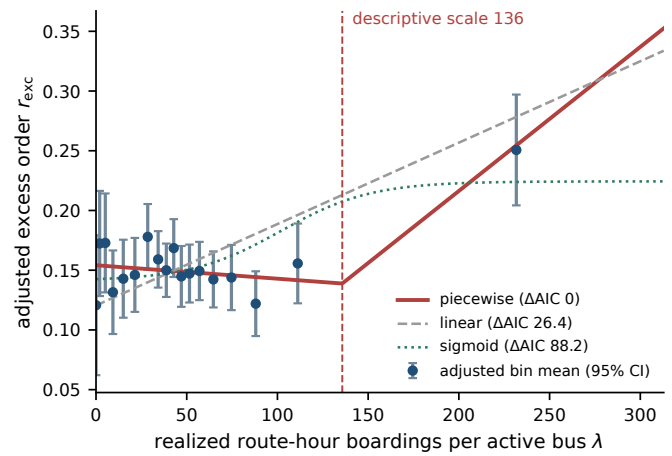


Fig. 5. Adjusted corrected order versus realized load. Points are equal-count-bin means of the piecewise-model partial residual with 95% normal intervals; curves hold line-direction and non-load controls at their sample-average values. AIC favors a descriptive change in slope near 136, but a breakpoint is not a critical point.

Table 3. Criticality decision gate.

Signature	Result	Pass
Interior susceptibility peak	absent	no
Near-scale bimodality	weaker than off-scale	no
Scale stable across N	$ r = 0.282$	yes
Free-exponent collapse	not identified	not scored

ent estimand because both the phase distribution and fleet size change.

D. Mechanisms: amplification and common shocks

Headway CV rises downstream across the 10–90% interior: the mean slope is 0.1281 ($p = 1.32 \times 10^{-68}$). In 5,049 cells with complete driver data, inherited initial CV is the dominant standardized predictor of remaining growth (-0.2917); the negative sign is a ceiling effect because already-irregular cells have less room to worsen. Realized load is much smaller (0.0149, CI [0.0018, 0.0280]); route length, stop density, and sinuosity are unresolved.

The empirical Newell parameter $k = \text{mean dwell per stop divided by mean headway}$ has median 0.0477. Across 5,900 cells, median measured growth is 0.0139 per stop, compared with fastest-mode prediction 0.0911 and mode-1 prediction 0.0162. Seventy-one percent of cells grow, but predicted and measured growth correlate only 0.0198 ($p = 0.1277$). The idealized theory gives a plausible scale and no-threshold mechanism but does not rank which cells amplify.

The GPS-derived dwell mediation indirect effect is 5.49×10^{-5} with bootstrap CI [$-0.000206, 0.000401$]. This is inconclusive, not evidence against passenger dwell: tickets are not linked to individual vehicles and GPS dwell is noisy.

For overlapping route pairs, shared-trunk minus off-trunk correlation is 0.05325 (95% CI [0.02444, 0.08207]). After both line series are residualized on shared-segment mean speed, the gap falls to 0.01273 ([$-0.01808, 0.04889$]), retaining 24%. Only 5.88% of pairs have significant nonzero-lag peaks, and transfer-

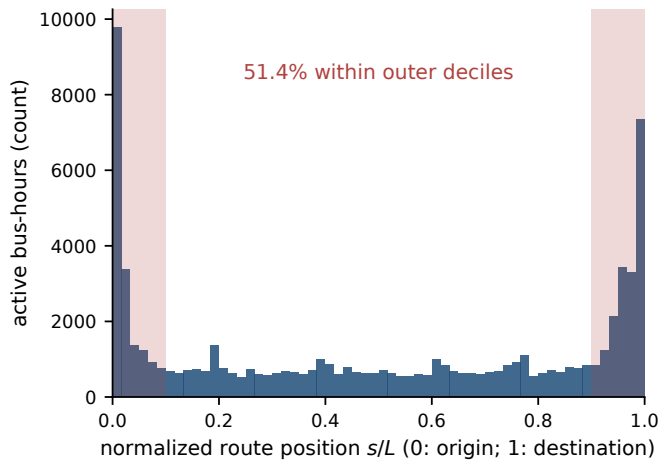


Fig. 6. Distribution of representative bus-hour positions. The shaded outer deciles contain 51.37% of active bus-hours after the layover rule. The plot establishes occupancy, not the cause of terminal waiting.

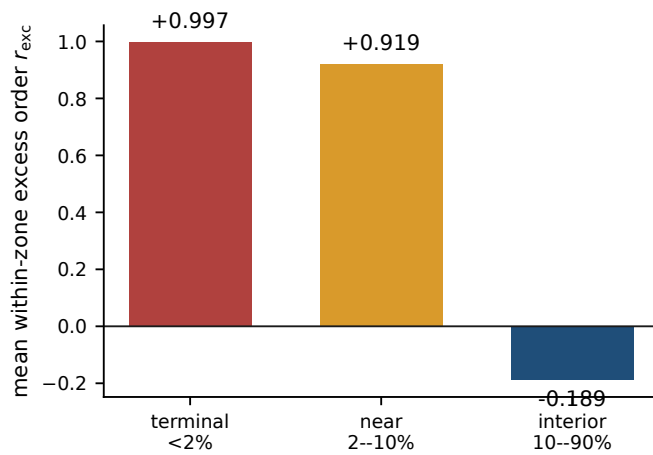


Fig. 7. Corrected order changes sign by route zone. Terminal concentration is nearly perfect by geometry, while interior vehicles are more evenly spaced than the reference null. These are zone-specific estimands because N is recomputed within each zone.

entropy directional asymmetry is null ($p = 0.652$). The evidence is largely consistent with common road-state exposure; speed control could also absorb part of a genuine coupling channel.

7. DISCUSSION

The main result is a measurement result with physical consequences. Raw r_1 is high in small fleets even without coordination, so conditions that change N inherit a false coherence contrast. Same- N correction is necessary, but it is not sufficient: the corrected statistic still depends on the phase coordinate, the occupancy null, and where buses sit on an open route. PPC reduces finite-sample bias under alternatives, while explicit null and phase sensitivity reveal which assumptions drive interpretation.

The data support a smooth operational crossover, not a demonstrated critical transition. AIC preference for a piecewise curve identifies a useful descriptive scale, but susceptibility and coexistence fail, and free-exponent scaling is not identified. This

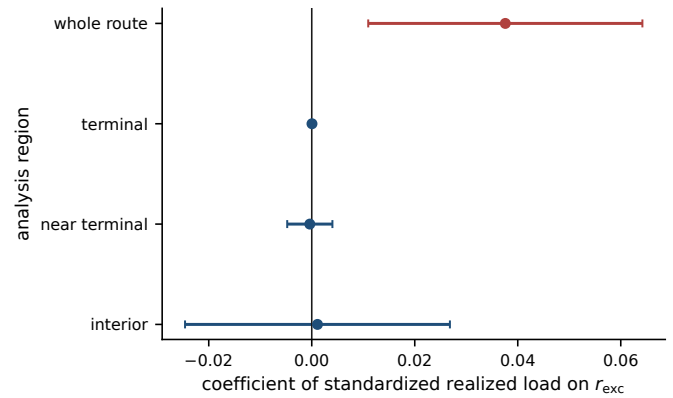


Fig. 8. Standardized load coefficients with line-direction fixed effects, log N control, and clustered 95% intervals. The association is detected for the whole-route mixture but not within terminal, near-terminal, or interior zones. Absence of detection is not proof of exact zero.

Table 4. Corrected order and load association by route zone.

Zone	Cells	mean r_{exc}	load coef.
Terminal < 2%	5,044	0.9974	0.00004
Near 2–10%	3,980	0.9186	−0.00038
Interior 10–90%	7,738	−0.1887	0.00111
Whole route	10,135	0.1574	0.03758

result is physically coherent: the appropriate nearest-neighbour theory is unstable for every positive k , whereas mean-field Kuramoto assumptions are violated by directed coupling, terminal resets, varying N , and frequency heterogeneity.

Spatial mixture is the strongest empirical explanation for apparent whole-route synchronization. The result is not merely that terminals contain many buses; it is that terminal and interior order have opposite signs, zone-specific load effects are absent, and direct terminal-fraction control substantially attenuates the whole-route coefficient. The residual coefficient and changed within-zone N prevent a claim of complete mediation.

Operationally, inherited irregularity and shared road conditions are more informative than a single passenger-load threshold. Interventions aimed at headway regularization and common corridor conditions are more directly supported than policies based on an inferred critical coupling.

8. LIMITATIONS

The missing vehicle crosswalk confines load to route-hour resolution and makes vehicle-level dwell mediation unidentifiable. Realized load divides by endogenous supply. Representative midpoint positions compress within-hour trajectories, while zone restriction and trimming change both occupancy and N . Nulls assume conditional exchangeability even though physical interactions are directed. Speed residualization may remove both common shocks and part of genuine coupling. The scaling objective has three free parameters, five strata, discontinuous bin membership, and boundary solutions. Finally, the record covers 19 days in one city; replication should test whether the large terminal mass is specific to this network.

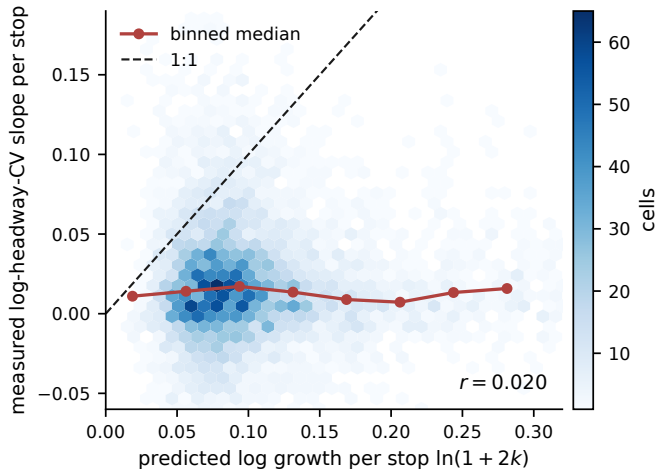


Fig. 9. Measured versus Newell–Potts fastest-mode growth. Binned medians remain nearly flat and the cell-level correlation is 0.020. Agreement of an average order of magnitude must not be mistaken for predictive validation.

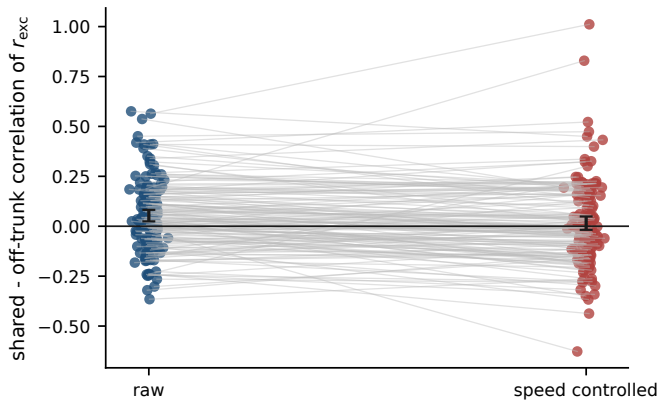


Fig. 10. Within-pair shared-minus-off-trunk correlations before and after controlling shared speed. Grey lines preserve pair identity; black intervals summarize the pair distribution. The controlled mean spans zero, supporting a largely common-shock interpretation.

9. CONCLUSION

The Niterói data can be made to look Kuramoto-like with a conventional order parameter, but that appearance does not identify critical synchronization. Exact finite-size reasoning changes the baseline; marginal and fleet-size-conditional weekend comparisons answer different questions; route-position analysis reveals a terminal/interior mixture; and common-speed control removes most shared-corridor coherence. The surviving physical picture is an unconditional, smoothly load-modulated operational instability embedded in boundary occupancy and common road shocks, not a demonstrated city-scale critical transition.

REPRODUCIBILITY STATEMENT

All numbers, tables, and independent figures in this report were regenerated from the released raw data in one corrected execution on 21 August 2026. The run uses continuous route geometry, explicit input/output paths, fixed random seeds for inferential

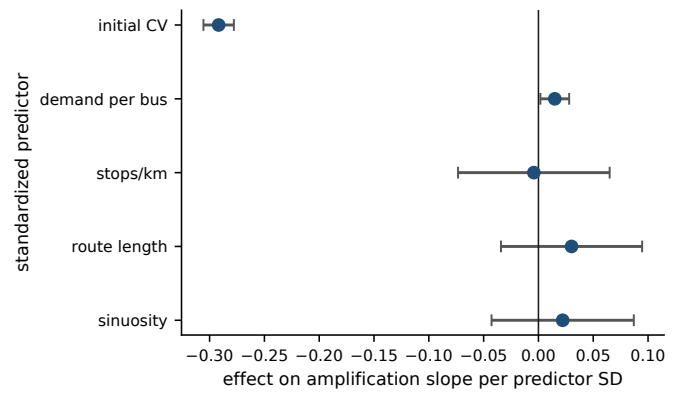


Fig. 11. Standardized predictors of downstream amplification with 95% intervals. The large negative initial-CV coefficient represents limited remaining growth, not a stabilizing effect of irregularity.

simulations, 26 passing unit tests, seven phase-null specifications, and an explicit non-identification gate for scaling collapse.

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